I decided to use some images of a castle that I found online. I cropped the 3 individual images by hand. And used the first and the second image to develop my image stitching pipeline. I started by implementing my pipeline with the idea of using all three images, but when I started to face issues I only kept 2 as it was easier to debug.

image1

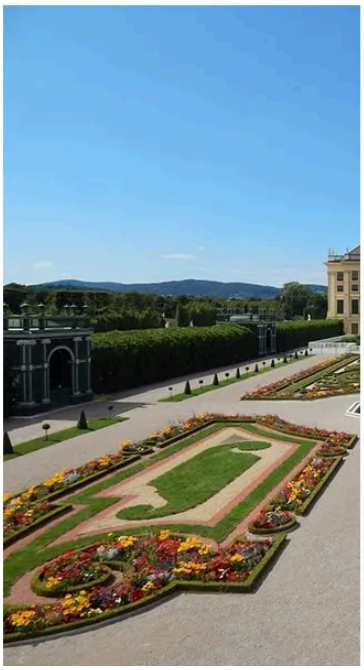


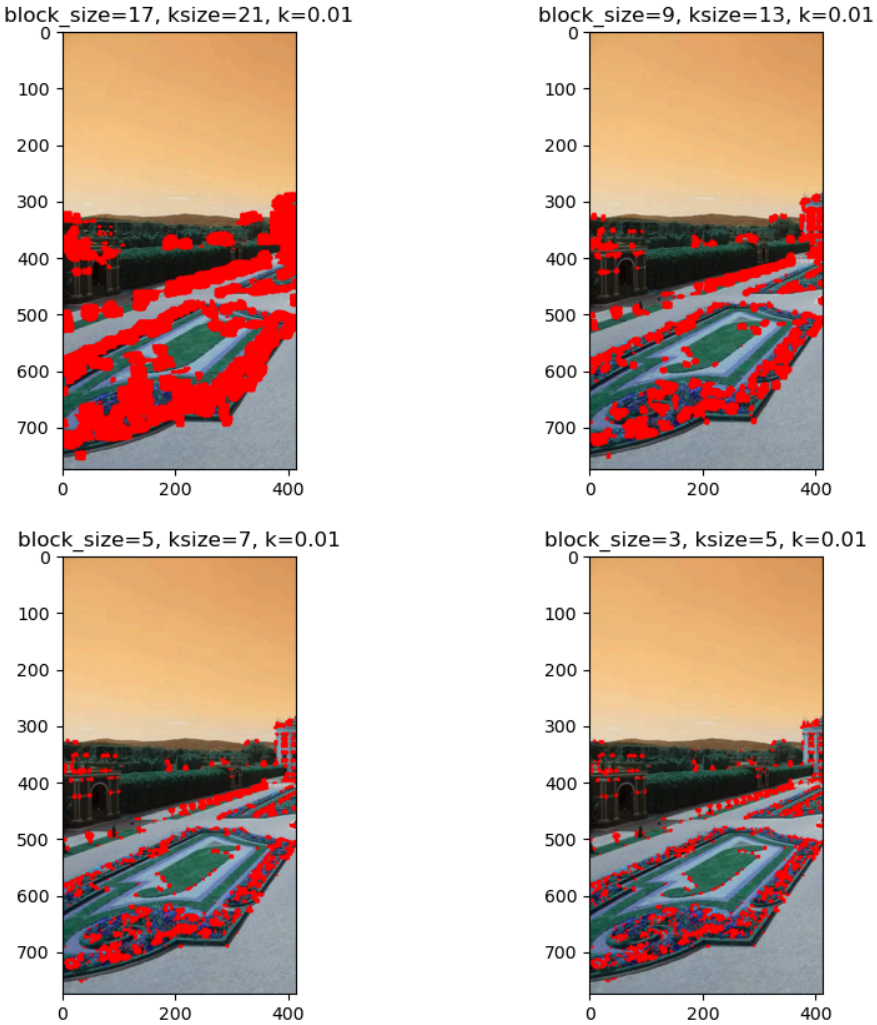
image2

image3

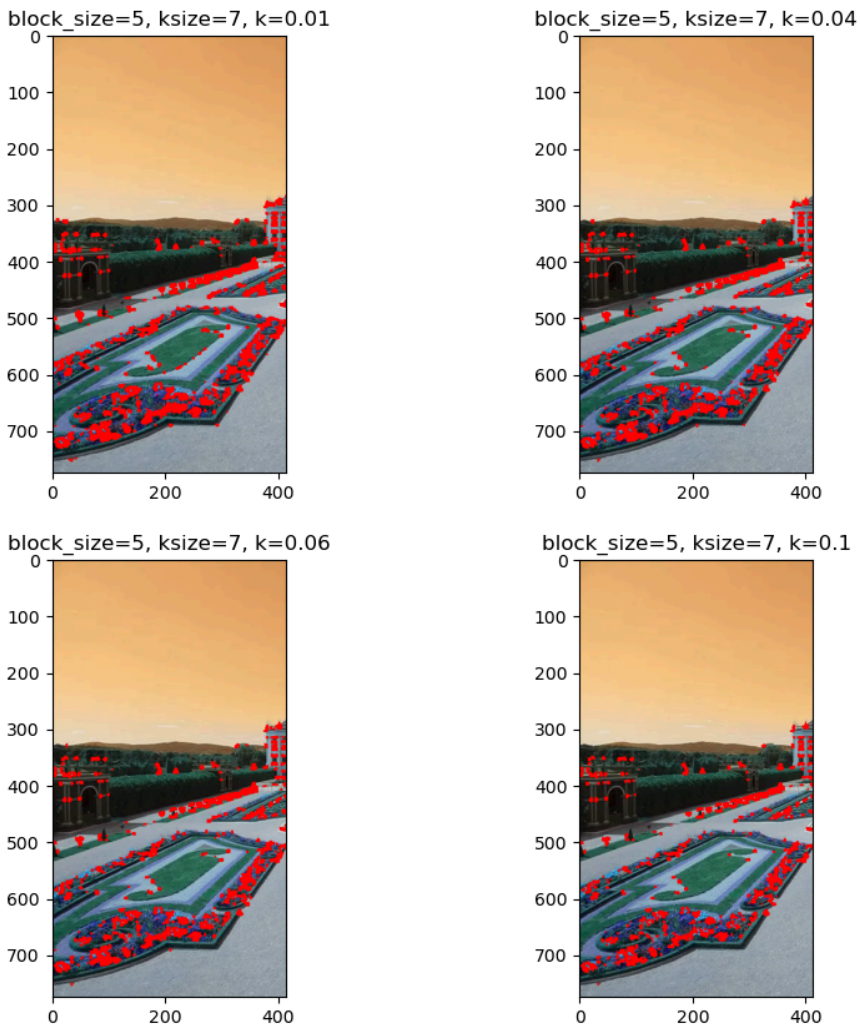
Harris Corner Detection:

To pick the parameter for my Harris detector, I ran it on the left image with different values. I first tested the size of the filters going from (17, 21), (9, 13), (5, 7), (3, 5). I picked these values arbitrarily. As we can see in the figure below, the smaller the values the more precise the points

seem to be, so in this case I decided to go with size (3,5).



I then played around with the parameter K, using the best value for my block size and ksize based on the experiment above. I tried values of 0.01,0.04, 0.06 and 0.1.



0.04 seems to be the best value in this case. We can see a lot of noise while using 0.01 but at 0.06 some corners were not picked anymore by the algorithm. This is why I settled for 0.04.

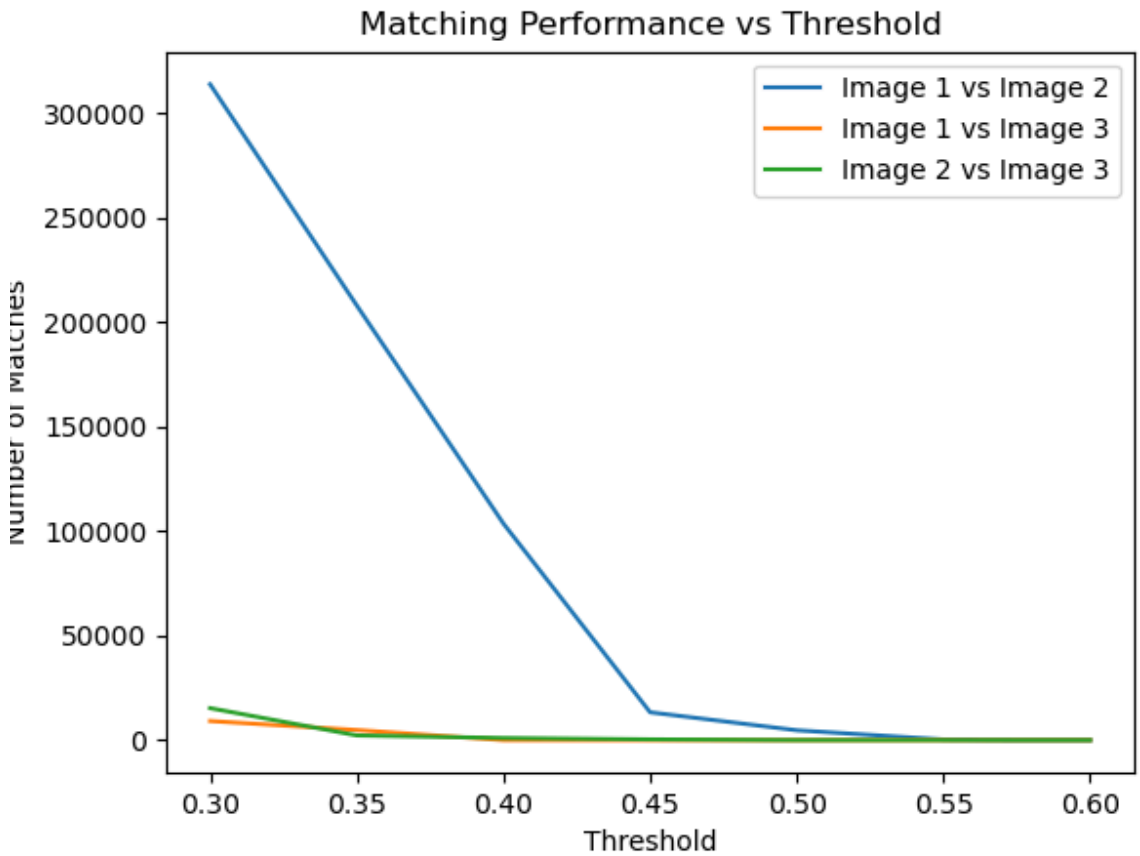
Sift Features:

After computing my key points I realized that I had way too many points. Given that in the future, I needed only a few good key points, I decided to keep only the 1500 best key points out of the ~ 20 000 ones. Reducing the number of key points made the next steps a lot faster to compute.

Distance between correlation and euclidean distance:

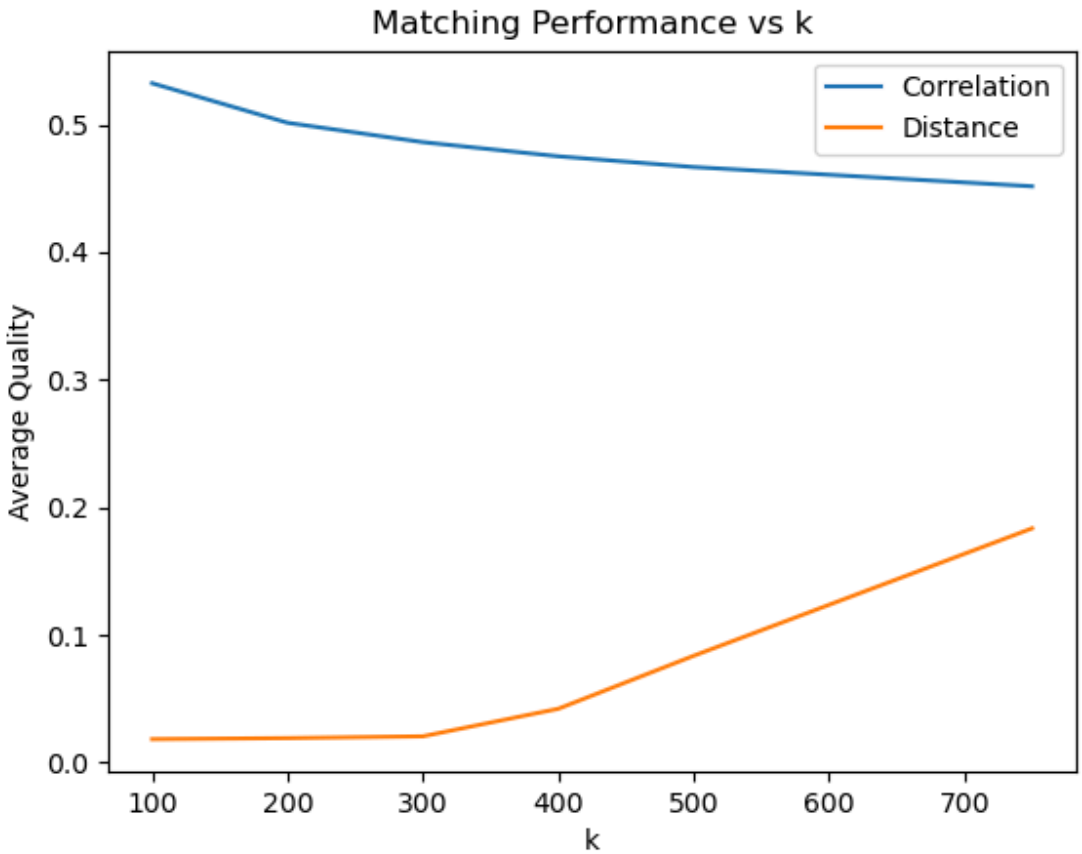
From the descriptors generated in the previous step, I was able to compute the correlation and euclidean distance without problem. From those correlations and distances I identified the best

key points by first using a threshold on the correlation. However, I found out that each image required its own tailored threshold to make sure we always get a similar amount of key points for each image. Meaning the threshold is not a reliable method. As shown in this figure:



Instead of using thresholds, I tried to keep only the top K key points which worked a lot better. I had to use a different method for keeping the top k points from the correlation set to the euclidean distance one. Regarding correlation, you want to keep the ones with the highest correlation but for distance you want the key points that have the smallest euclidean distance.

I did some experimentation with the average quality of the subset when taking different values for the number of samples in the subset. As shown in the plot below, the quality in the distance gets quite bad after 400, with a small increase after 300. I started by taking 300 key points, but realized that my RANSAC would take all the inliers, probably because they were already filtered so much. So I increased it to 400.



RANSAC

Regarding RANSAC, during the implementation I had an issue where my inlier threshold was too high, meaning that a lot of the points were being classified as inliers and I ended up with around 80% of my key points being classified as inliers. However, now that I fixed my issue, I still get a lot of inliers, but for a different reason. I have filtered the key points a lot such that in the end most of them are good ones, even with a threshold of 0.03 I get only 26 outliers.

When computing the average residuals of the outliers, I get a value of 4.27. Again this is due to the fact that my key points have been filtered a lot already and a lot of good ones are remaining. If I had more key points I would expect the residuals to increase.

In the following figure, we can see the selected key points in both images.

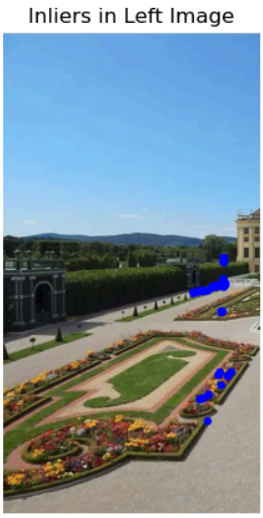
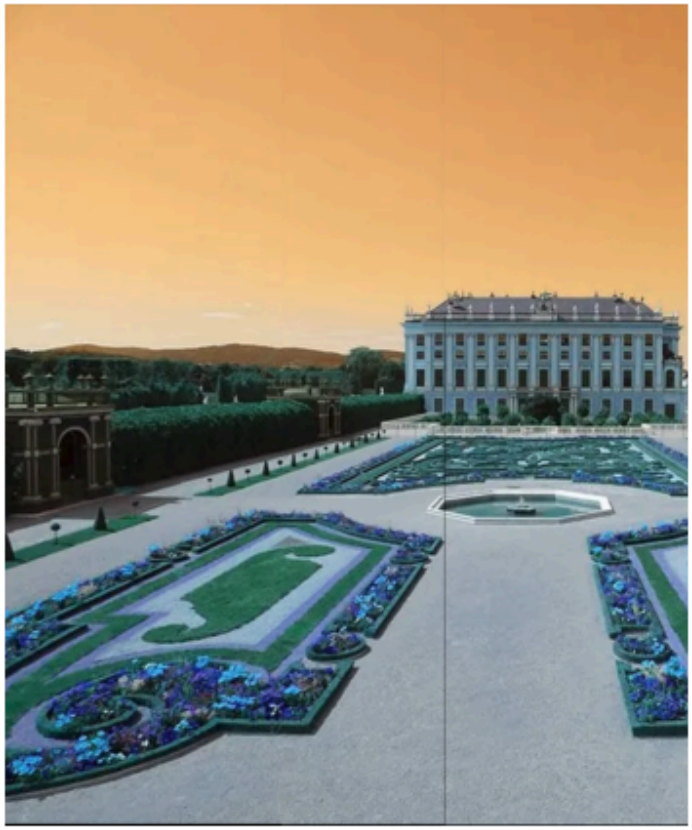


Image stitching

Combining the images is the task that was the most challenging for me, as I didn't really understand how to merge them together. And even now, the resulting image has a horizontal line in the middle, so I must have done something wrong.